

Notes: Olken and Barron (JPE, 2009)

The Simple Economics of Extortion: Evidence from Trucking in Aceh

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1 Big picture

- **Research question.** Do corrupt officials behave like firms in a market? More specifically, do police, military officers, and weigh-station officials set illegal payments in ways that are consistent with industrial organization models of market structure, bargaining, and price discrimination?
- **Main idea.** Corruption is not just a set of isolated transactions. When a truck must pass many corrupt checkpoints, each official is one seller in a chain. The number and location of these sellers can affect the bribe each one charges.
- **Empirical setting.** The paper studies trucks traveling to and from Aceh, Indonesia. Surveyors accompanied truck drivers on 304 trips and directly observed more than 6,000 illegal payments to police, soldiers, and weigh-station attendants.
- **Key shock.** After the 2005 Aceh peace agreement, the Indonesian government withdrew many police and military personnel from Aceh. This reduced the number of checkpoints in Aceh but not in neighboring North Sumatra.
- **Main finding on market structure.** As checkpoints in Aceh disappeared, average bribes at remaining checkpoints in North Sumatra increased. This is exactly the kind of endogenous price response predicted by models of sequential monopoly and double marginalization.
- **Main finding on bargaining.** Bribes are not completely posted prices. Observable bargaining power matters: officers with visible guns and checkpoints with more officers charge higher bribes and are more likely to negotiate.
- **Main finding on holdup.** On the Meulaboh route, checkpoints closer to the end of the trip collect higher payments. This is consistent with downstream holdup: late checkpoints can extract more surplus because the driver has already sunk most of the trip cost.
- **Main finding on price discrimination.** Corrupt officials charge different prices to different trucks. Newer trucks, higher-value cargo, and some cargo types pay more. At one weigh station, officials use a coupon system that resembles a menu of two-part tariffs.
- **Economic takeaway.** The illegal nature of corruption does not eliminate market forces. Corrupt officials appear to respond to incentives, market structure, bargaining power, and opportunities for price discrimination.

2 Incremental contribution of the paper

- **From corruption as crime to corruption as pricing.** Many studies ask whether officials take bribes. Olken and Barron ask whether bribe setting follows standard pricing logic from industrial organization.
- **Direct observation of bribes.** Instead of relying on retrospective reports, the authors directly observe thousands of payments during actual truck trips. This greatly improves measurement of locations, amounts, and transaction details.
- **A real market-structure shock.** The withdrawal of troops and police from Aceh changes the number of checkpoints. This makes it possible to estimate how bribe prices respond when the number of corrupt sellers changes.

- **Testing Shleifer–Vishny-style corruption theory.** The paper directly tests the idea that decentralized corruption can generate double marginalization, where each corrupt seller ignores the effect of his price on the total cost of completing the route.
- **Micro evidence on bargaining.** The paper observes whether guns are visible, how many officers are present, and whether negotiation occurs. This lets the authors test whether payments are fixed in advance or partly determined by ex post bargaining.
- **Evidence on sophisticated illegal contracts.** The coupon system at the Gebang weigh station shows that even illegal markets can sustain nonlinear pricing arrangements similar to those studied in standard monopoly theory.
- **Policy relevance.** The paper highlights a subtle anticorruption point: removing some bribe takers may reduce total bribes less than expected if remaining bribe takers raise their prices.

3 Setting and data

3.1 Trucking routes in Aceh

The study focuses on two long-distance trucking routes linking Aceh to Medan, the capital of North Sumatra. The first route runs from Meulaboh to Medan; the second runs from Banda Aceh to Medan. Both routes contain portions inside Aceh and portions inside North Sumatra.

- **Meulaboh–Medan route.** About 637 kilometers; a typical truck takes roughly 35 hours.
- **Banda Aceh–Medan route.** About 560 kilometers; a typical truck takes roughly 24 hours.
- **Checkpoints.** Police and military checkpoints stop trucks and demand payments. These payments look more like tolls or extortion than bribes to avoid a specific violation.
- **Weigh stations.** Trucks are often overweight. Official penalties are rarely imposed; instead, drivers pay informal payments to continue.
- **Other illegal payments.** Some trucks pay convoy fees or monthly protection fees to criminal organizations or security actors.

Interpretation. The road network creates a chain of required approvals. A truck must pass all relevant checkpoints and weigh stations for the trip to succeed. This makes the setting naturally suited to a model of sequential monopolists.

3.2 Military presence and withdrawal

Aceh had a long-running conflict between the Indonesian government and the Free Aceh Movement. After the August 2005 peace agreement, many police and military personnel were withdrawn from Aceh.

- The withdrawal occurred in waves from late 2005 into early 2006.
- The withdrawal affected personnel stationed in Aceh, not personnel in North Sumatra.

- Since many checkpoints were staffed by these personnel, the withdrawal reduced the number of checkpoints on the Aceh portions of the routes.
- This provides a source of plausibly exogenous variation in the “market structure” for bribe collection.

Interpretation. The paper’s central identification strategy compares how payments in North Sumatra respond to reductions in checkpoints in Aceh. North Sumatra checkpoints themselves are not directly affected by the withdrawal, so changes in their prices reveal strategic responses to the disappearance of other checkpoints along the route.

3.3 Data collection

The authors collected data by hiring local surveyors to accompany truck drivers on regular trips.

- Surveyors recorded the time, location, and amount of each illegal payment.
- They recorded whether the checkpoint was run by police, army, or another organization.
- They recorded whether officers were visibly armed and how many officers were visible.
- They recorded truck characteristics, cargo type, cargo weight, and driver characteristics.
- The sample is not a random sample of all trucks, but it is close to a within-firm sample of regular trips by cooperating firms.

Measurement advantage. Direct observation avoids two major problems in survey data: drivers may not remember all checkpoints precisely, and they may exaggerate bribe amounts when asked after the fact.

3.4 Descriptive facts

- The average one-way trip cost is roughly Rp. 2.9 million, about US\$325 at the exchange rate used in the paper.
- Illegal payments average about Rp. 361,000 per trip, or about 13% of total trip costs.
- The average trip encounters about 20 checkpoints.
- Meulaboh trips face many more checkpoints than Banda Aceh trips: about 27 versus 11 on average.
- Checkpoint payments are much larger on the Meulaboh route, while weigh-station payments are more important on the Banda Aceh route.

Interpretation. Illegal payments are economically meaningful. They are not tiny nuisance costs; they are comparable to, and in some cases larger than, the wages paid to the truck driver and assistant.

4 Theory and empirical specifications

4.1 Market structure and double marginalization

The paper begins with a simple model in which checkpoints are arranged along a route and each checkpoint charges a price for passage.

Let there be n checkpoints and let the total price to complete the route be P . If each checkpoint sets its own price independently, each checkpoint ignores the effect of its price on the revenues of the other checkpoints. The equilibrium condition implies:

$$\frac{q'(P)P}{q(P)} = -n.$$

- When there are more checkpoints, the total route price tends to be higher.
- When there are more checkpoints, the average price per checkpoint tends to be lower.
- Therefore, when some checkpoints are removed, remaining checkpoints should raise their prices.

If instead a central authority sets all checkpoint prices jointly, the total monopoly price is independent of the number of checkpoints. In that case, removing checkpoints would cause the remaining checkpoints to raise prices enough to keep total payments constant.

Empirical prediction. If prices respond to market structure, the elasticity of the average bribe at remaining checkpoints with respect to the total number of checkpoints should be negative.

4.2 North Sumatra response specification

The main checkpoint-level regression is:

$$\log(\text{Price}_{ci}) = \alpha_c + X_i' \gamma + \beta \log(\text{ExpectedPosts}_i) + \varepsilon_{ci},$$

where c indexes checkpoints and i indexes trips. The dependent variable is the log bribe paid at a North Sumatra checkpoint. The key variable is the log expected number of checkpoints on the full route.

- The expected number of checkpoints is calculated from other trips in a two-week window.
- North Sumatra checkpoint counts are held constant in constructing the variable, so the identifying variation comes from changes in Aceh.
- Controls include cargo, driver salary, truck age, truck age squared, and tons overweight.
- The specification includes checkpoint-by-direction fixed effects.

The trip-level version is:

$$\log(\text{Payments}_i) = \alpha + X_i' \gamma + \beta \log(\text{ExpectedPosts}_i) + \varepsilon_i,$$

where the dependent variable is total illegal payments in North Sumatra on trip i .

Interpretation of β .

- $\beta = 0$: prices do not respond to market structure.
- $\beta = -1$: remaining checkpoints fully offset the lost checkpoints, consistent with centralized pricing.
- $-1 < \beta < 0$: prices respond, but not fully; this is consistent with decentralized pricing.

4.3 Within-district specification

The second empirical strategy looks within Aceh districts and uses the fact that troop withdrawals occurred at different times in different districts. The regression is:

$$\log(\text{Payments}_{di}) = \alpha_i + \alpha_d + \beta \log(\text{ExpectedPosts}_{di}) + \varepsilon_{di},$$

where d indexes districts and i indexes trips.

- The dependent variable is total payments made in district d on trip i .
- Trip fixed effects absorb all trip-level demand and route conditions.
- District-by-direction fixed effects absorb permanent district differences.
- The coefficient asks whether payments in a district fall proportionally when checkpoints in that district disappear.

Interpretation.

- $\beta = 1$: no price response; payments fall one-for-one with checkpoints.
- $\beta = 0$: full offset within district, consistent with centralized district pricing.
- $0 < \beta < 1$: partial offset, consistent with decentralized price adjustment.

4.4 Bargaining and sequential holdup

The paper then asks whether checkpoint payments are posted prices or are partly bargained. The bargaining-power regression is:

$$\log(\text{Price}_{ci}) = \alpha_i + \alpha_c + \delta_1 \text{Gun}_{ci} + \delta_2 \text{NumOfficers}_{ci} + \varepsilon_{ci}.$$

- A visible gun should increase the officer's bargaining power.
- More visible officers should also increase the checkpoint's bargaining power.
- If these variables raise prices and negotiation probability, that supports a bargaining interpretation.

Given bargaining, downstream checkpoints should be able to extract more surplus. The paper tests this with:

$$\log(\text{Price}_{ci}) = \alpha_i + \alpha_c + \beta \text{MeanPercentile}_{ci} + \varepsilon_{ci},$$

where $\text{MeanPercentile}_{ci}$ measures how far along the trip the checkpoint is, from 0 at the beginning to 1 at the end.

Prediction. If holdup is important, $\beta > 0$: prices should rise as the truck gets closer to its destination.

4.5 Price discrimination

Finally, the paper asks whether corrupt officials can price discriminate. The main regression is:

$$\log(\text{Price}_{ci}) = \alpha_c + X_i' \beta + \varepsilon_{ci},$$

where X_i includes observable characteristics of the truck, cargo, and driver.

- Higher-value cargo should be associated with greater willingness to pay.
- Newer trucks and higher-income drivers may be charged more.
- More frequent drivers may pay less because they know the going price better.
- At Gebang, the paper also studies a coupon system that creates a menu of two-part tariffs.

5 Identification and interpretation

5.1 What the paper can identify

The strongest evidence concerns how bribe prices respond to changes in the number of checkpoints. Because the troop withdrawal changed checkpoint availability in Aceh while North Sumatra checkpoints remained in place, the authors can test whether remaining bribe takers change prices when other bribe takers disappear.

5.2 Why the withdrawal is useful

The withdrawal is useful because it is driven by the peace agreement rather than by ordinary trucking demand. The timing and location of personnel reductions provide variation in checkpoint density that is plausibly external to any one driver's willingness to pay.

5.3 Route comparison

The Meulaboh route experienced a much larger checkpoint decline than the Banda Aceh route. This allows the paper to estimate difference-in-difference specifications using both routes, controlling for common time trends with either a cubic time trend or month fixed effects.

5.4 Direct observation and Hawthorne effects

The data collection method is unusually strong because surveyors directly observe payments. One concern is that drivers might behave differently because a surveyor is present. The paper argues that this is unlikely to drive the main results because drivers already have an incentive to minimize bribe payments, and any observer effect should mostly affect levels rather than changes across time, checkpoints, or routes.

5.5 Interpretation of the elasticities

The key elasticities are reduced-form measures of price response. They should not be interpreted as structural estimates of a full corruption model. A negative elasticity means that remaining checkpoint officials raise prices when the total number of checkpoints falls.

5.6 Limitations

- The truck sample is not a random sample of all trucks in Aceh.
- The paper studies one conflict-affected region and one trucking market.
- The withdrawal shock is plausibly exogenous, but other time-varying conditions may still matter.
- The estimates cannot always distinguish sharply between centralized and decentralized bribe setting.

Interpretation. The paper is best read as strong micro evidence that corrupt pricing follows market incentives, rather than as a universal estimate of corruption market structure in all settings.

6 Tables and figures

6.1 Figure 1 and Table 1: Routes and summary statistics

Read Figure 1:

- Figure 1 maps the two trucking routes: Meulaboh–Medan and Banda Aceh–Medan.
- Both routes cross Aceh and North Sumatra, which is essential for the identification strategy.
- The map also marks the weigh stations, including Gebang, Seumedam, Doulu, and Sidikalang.

Read Table 1:

- Table 1 reports trip expenditures and illegal payments by route.
- Total trip expenditures average about Rp. 2.9 million.
- Bribes, extortion, and protection payments average Rp. 361,323 across both roads.
- Checkpoint payments are much larger on the Meulaboh route, while weigh-station payments are larger on the Banda Aceh route.

- The average number of checkpoints is 20 overall, 27 on the Meulaboh route, and 11 on the Banda Aceh route.

Economic message: Illegal payments are a large component of transport costs, and the two routes differ sharply in the composition of corruption payments. This heterogeneity helps the paper test market-structure predictions.

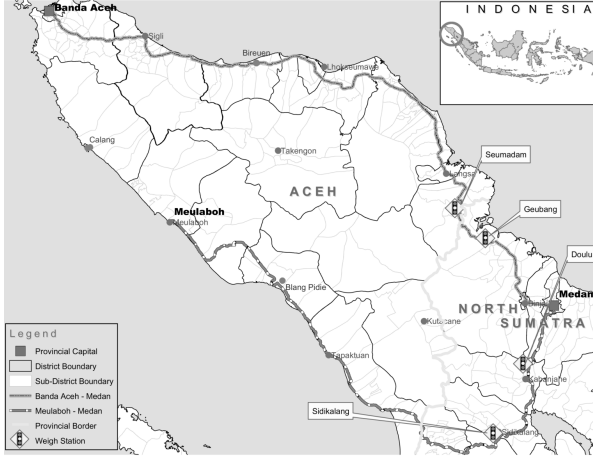


Figure 1: Routes.

TABLE 1
SUMMARY STATISTICS

	Both Roads (1)	Meulaboh Road (2)	Banda Aceh Road (3)
Total expenditures during trip (rupiah)	2,901,345 (725,003)	2,932,687 (561,736)	2,863,637 (883,308)
Bribes, extortion, and protection payments	361,323 (182,563)	415,263 (180,928)	296,427 (162,896)
Payments at checkpoints	131,876 (106,386)	201,671 (85,203)	47,905 (57,293)
Payments at weigh stations	79,195 (79,405)	61,461 (43,090)	100,531 (104,277)
Convoy fees	131,404 (176,689)	152,131 (147,927)	106,468 (203,875)
Coupons/protection fees	18,848 (57,593)	...	41,524 (79,937)
Fuel	1,553,712 (477,207)	1,434,608 (222,493)	1,697,010 (637,442)
Salary for truck driver and assistant	275,058 (124,685)	325,514 (139,233)	214,353 (65,132)
Loading and unloading of cargo	421,408 (336,904)	471,182 (298,246)	361,523 (370,621)
Food, lodging, etc.	148,872 (70,807)	124,649 (59,067)	178,016 (72,956)
Other	140,971 (194,728)	161,471 (236,202)	116,308 (124,755)
Number of checkpoints	20 (13)	27 (12)	11 (6)
Average payment at checkpoint	6,262 (3,809)	7,769 (1,780)	4,421 (4,722)
Number of trips	282	154	128

NOTE.—Standard deviations are in parentheses. Summary statistics include only those trips for which salary information was available. All figures are in October 2006 rupiah (US\$1.00 = Rp. 9,200).

Table 1: Summary statistics.

6.2 Figure 2 and Figure 3: Weigh-station pricing and troop withdrawal

Read Figure 2:

- Figure 2 plots bribes at each weigh station against the number of tons the truck is overweight.
- At all four weigh stations, payments increase with excess weight.
- The positive slope means weigh-station corruption still creates some marginal cost of being overweight.
- The schedule is much flatter than the official penalty schedule and likely flatter than the social marginal damage from overweight trucks.

Read Figure 3:

- The left panels show checkpoints in Aceh falling sharply on the Meulaboh route while North Sumatra checkpoints remain roughly stable.
- The middle panels show average bribes at North Sumatra checkpoints.
- On the Meulaboh route, average North Sumatra bribes rise as Aceh checkpoints disappear.
- The Banda Aceh route shows a much weaker pattern because most Aceh checkpoints there had already disappeared before data collection began.

Economic message: The visual evidence already suggests endogenous price adjustment. Removing checkpoints does not mechanically reduce bribes one-for-one because remaining checkpoint officials raise prices.

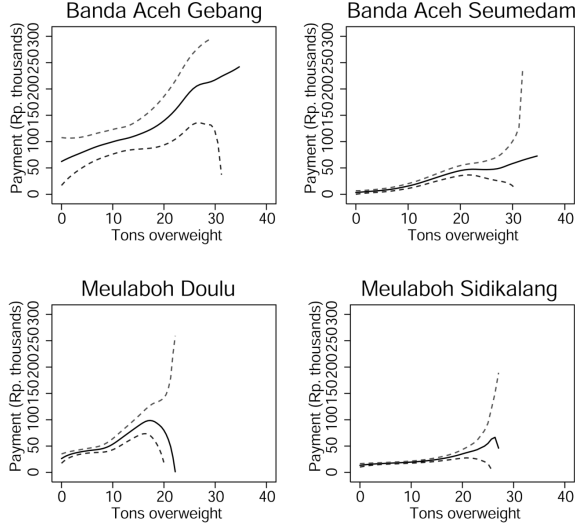


Figure 2: Payments at weigh stations.

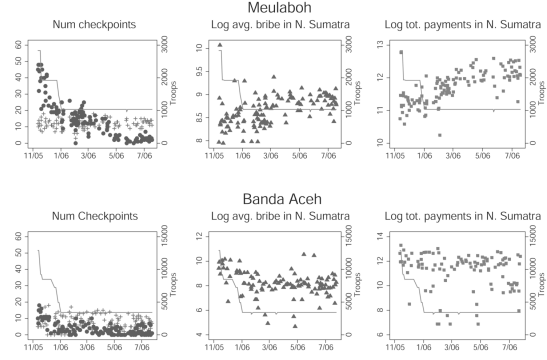


Figure 3: Impact of troop withdrawals.

6.3 Tables 2 and 3: Market structure and checkpoint prices

Read Table 2:

- Table 2 estimates how bribes in North Sumatra respond to changes in expected checkpoints on the route.
- In panel A, the elasticity of the average checkpoint payment is negative across all specifications.
- The Meulaboh OLS estimates are about -0.55 to -0.58 , and the IV estimate is about -0.79 .
- Panel B shows similar results for total illegal payments in North Sumatra.
- The estimates are statistically different from zero, so prices clearly respond to market structure.
- In several specifications the estimates are not statistically distinguishable from -1 , so the table cannot always separately identify centralized versus decentralized pricing.

Read Table 3:

- Table 3 estimates a district-level specification within Aceh using trip fixed effects.
- The OLS elasticities are 0.663 on the Meulaboh route and 0.586 using both routes.
- These estimates are positive but significantly below one.
- This means that payments in a district fall when checkpoints fall, but less than mechanically one-for-one.
- The evidence points to partial price adjustment within districts.

Economic message: Both strategies show that bribe prices are endogenous to market structure. Corruption markets look like economic markets: when some sellers disappear, remaining sellers raise prices.

TABLE 2
IMPACT OF NUMBER OF CHECKPOINTS IN ACEH ON BRIBES IN NORTH SUMATRA

	Meulaboh OLS (1)	Meulaboh OLS (2)	Meulaboh (Pre-Press Conference) OLS (3)	Meulaboh IV (4)	Both Routes OLS (5)	Both Routes OLS (6)
A. Log Payment at Checkpoint						
Log expected checkpoints on route	-.545*** (.157)	-.580*** (.167)	-.684*** (.257)	-.788*** (.217)	-.701*** (.202)	-.787*** (.203)
Truck controls	No	Yes	Yes	Yes	Yes	Yes
Common time effects	None	None	None	None	Cubic	Month FE
Observations	1,941	1,720	1,069	1,720	2,369	2,369
Test elasticity = 0	.00	.00	.01	.00	.00	.00
Test elasticity = -1	.00	.01	.22	.33	.14	.29
B. Log Total Payments						
Log expected checkpoints on route	-.736*** (.064)	-.695*** (.069)	-.643*** (.237)	-.782*** (.131)	-1.107** (.444)	-1.026** (.405)
Truck controls	No	Yes	Yes	Yes	Yes	Yes
Common time effects	None	None	None	None	Cubic	Month FE
Observations	161	144	90	144	249	249
Test elasticity = 0	.00	.00	.01	.00	.01	.01
Test elasticity = -1	.00	.00	.14	.10	.81	.95

NOTE.—Panel A presents the results from estimating eq. (5), where each observation is a payment at a checkpoint, the dependent variable is the log payment at the checkpoint, the sample is limited to North Sumatra province only, all specifications include checkpoint $\times$ direction of travel fixed effects, and robust standard errors are in parentheses, adjusted simultaneously for clustering at the checkpoint and trip levels. Panel B presents the results from estimating eq. (6), where each observation is a trip, the dependent variable is log total payments in North Sumatra province, and robust Newey-West standard errors allowing for up to 10 lags are included in parentheses. In both specifications, truck controls are dummies for six types of contents, log driver's monthly salary, truck age and truck age squared, and number of tons truck is overweight; these characteristics are examined in more detail in table 6. The instrument in col. 4 is the log number of troops remaining in Aceh in the districts covered by the Meulaboh route; the first-stage F -statistic for the excluded instruments based on the panel B specification is 43.11. Log expected checkpoints uses only variation from Aceh province; the details of how this variable is constructed are in the text. Columns 5 and 6 are the difference-in-difference specifications, including both routes and a common cubic in time (col. 5) or common month fixed effects (col. 6). Note that cols. 5 and 6 of panel B also includes a route dummy.

* Significant at 10 percent.
** Significant at 5 percent.
*** Significant at 1 percent.

Table 2: Impact of number of checkpoints in Aceh on bribes in North Sumatra.

TABLE 3
IMPACT OF NUMBER OF CHECKPOINTS ON TOTAL PAYMENTS IN DISTRICT

	Meulaboh OLS (1)	Meulaboh IV (Troops) (2)	Both Routes OLS (3)	Both Routes IV (Troops) (4)
Log expected checkpoints in district	.663*** (.081)	1.522*** (.390)	.586*** (.082)	.786** (.359)
Observations	1,090	1,026	1,435	1,363
Test elasticity = 0	.00	.00	.00	.03
Test elasticity = 1	.00	.18	.00	.55

NOTE.—This table presents the results from estimating eq. (7), where there is one observation for each district encountered on each trip, the dependent variable is the log of total amount paid in bribes in the district, and trip fixed effects and district $\times$ direction of travel fixed effects are included. The instrument in cols. 2 and 4 is the log number of troops remaining in the district. Robust standard errors are in parentheses, adjusted simultaneously for clustering at the district $\times$ quarter and trip levels.

Table 3: Impact of number of checkpoints on total payments in district.

6.4 Table 4 and Figure 4: Bargaining and downstream holdup

Read Table 4:

- Table 4 tests whether bribes are partly determined by bargaining.
- A visible gun raises the log payment by about 0.15–0.17 and raises the probability of negotiation by about 4–5 percentage points.
- Each additional officer visible at the checkpoint raises the log payment by about 0.05 and raises the probability of negotiation by about 1.6–1.7 percentage points.
- Characteristics of the subsequent checkpoint do not predict the current bribe.
- This supports the interpretation that visible guns and more officers increase local bargaining power rather than simply proxying for general security conditions.

Read Figure 4:

- Figure 4 plots residualized checkpoint payments against the share of the trip completed.

- On the Meulaboh route, payments rise as the truck gets closer to the destination.
- On the Banda Aceh route, the pattern is weaker and nonmonotonic.
- The stronger Meulaboh pattern is consistent with sequential holdup along a route with fewer major intermediate destinations.

Economic message: Bribes are not just fixed posted prices. Bargaining power affects payments, and downstream checkpoints can exploit the fact that the driver has already incurred most of the trip cost.

TABLE 4
BARGAINING VERSUS FIXED PRICES

	LOG PAYMENT		NEGOTIATE DUMMY	
	(1)	(2)	(3)	(4)
Gun visible	.166** (.066)	.154** (.070)	.042** (.018)	.047*** (.018)
Gun visible at subsequent checkpoint		.016 (.024)		.016 (.018)
Number of officers at checkpoint	.047*** (.010)	.050*** (.009)	.017*** (.004)	.016*** (.005)
Number of officers at subsequent checkpoint		-.003 (.007)		-.003 (.004)
Observations	5,260	4,968	5,281	4,989
Mean dependent variable	8.49	8.50	.13	.13

NOTE.—This table presents the results from estimating eq. (8), where there is one observation for each payment at a checkpoint, and trip fixed effects and checkpoint $\times$ direction $\times$ month interval fixed effects are included. Robust standard errors are in parentheses, adjusted for clustering at the checkpoint level.

* Significant at 10 percent.

** Significant at 5 percent.

*** Significant at 1 percent.

Table 4: Bargaining versus fixed prices.

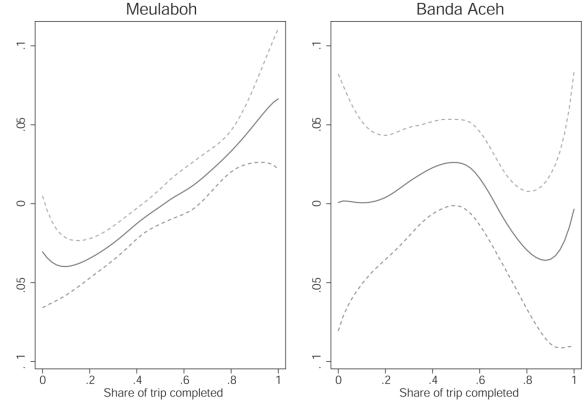


Figure 4: Payments by percentile of trip.

6.5 Table 5 and Figure 5: Sequential bargaining and observable price discrimination

Read Table 5:

- Table 5 gives the regression version of the sequential bargaining test.
- On the Meulaboh route, the coefficient on mean percentile is 0.145 and statistically significant.
- This implies payments increase from the beginning to the end of the trip.
- On the Banda Aceh route, the point estimate is negative and statistically imprecise.

Read Figure 5:

- Figure 5 shows nonparametric evidence of price discrimination based on observable characteristics.
- Older trucks, especially those above roughly 12 years old, pay lower prices.
- Trucks with very low cargo value per ton pay much lower bribes.
- Higher-value cargo is associated with higher checkpoint payments.

Economic message: The holdup result is strongest on the Meulaboh route. The price-discrimination evidence shows that corrupt officials use observable truck and cargo characteristics to infer willingness to pay.

TABLE 5
SEQUENTIAL BARGAINING AND INCREASING PRICES

	Meulaboh (1)	Banda Aceh (2)
Mean percentile	.145*** (.045)	-.178 (.225)
Observations	4,190	1,089

NOTE.—This table presents the results from estimating eq. (9), where there is one observation for each payment at a checkpoint, and trip fixed effects and checkpoint $\times$ month interval fixed effects are included. Robust standard errors are in parentheses, adjusted for clustering at the checkpoint level.

* Significant at 10 percent.

** Significant at 5 percent.

*** Significant at 1 percent.

Table 5: Sequential bargaining and increasing prices.

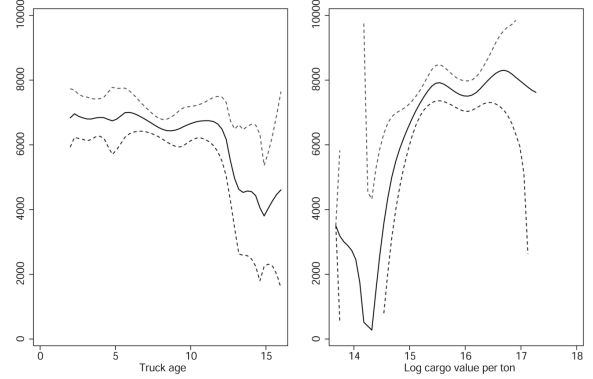


Figure 5: Price discrimination on observable characteristics.

6.6 Table 6 and Figure 6: Regression evidence and two-part tariffs

Read Table 6:

- Table 6 estimates checkpoint payments as a function of truck, cargo, and driver characteristics.
- Steel cargo is associated with substantially higher payments, while manufactured goods are associated with lower payments.
- Higher log cargo value per ton is positively associated with payments.
- Truck age is positive in column 1, though the nonlinear relationship is not always precisely estimated.
- Frequent drivers pay less, consistent with experience or knowledge of going prices.
- Higher driver salary is associated with higher payments in column 1.

Read Figure 6:

- Figure 6 shows the Gebang weigh-station coupon system.
- Trucks without a coupon face a lower fixed payment but a higher marginal cost when heavily overweight.
- Trucks with a coupon pay a higher fixed amount but a flatter schedule.
- The two schedules cross at roughly 16 tons overweight.

Economic message: Corruption is illegal, but illegal status does not prevent sophisticated pricing. The Gebang coupon resembles a two-part tariff menu designed to separate high- and low-willingness-to-pay trucks.

TABLE 6
PRICE DISCRIMINATION

	LOG PAYMENT	
	(1)	(2)
Contents of truck:		
Steel	.387*** (.144)	
Construction materials	-.063 (.059)	
Food	.025 (.039)	
Agricultural produce	.106 (.073)	
Manufactured goods	-.210*** (.074)	
Empty bottles	-.037 (.529)	
Log cargo value per ton		.072** (.036)
Truck characteristics:		
Truck age	.050** (.025)	.028 (.031)
Truck age squared (years)	-.002 (.002)	-.001 (.002)
Tons overweight	-.001 (.003)	.006** (.003)
Driver characteristics:		
Age	.006 (.004)	.001 (.004)
Years of education	.002 (.004)	-.011 (.007)
Speaks Acehnese	-.029 (.039)	.045 (.045)
Years of experience	-.003 (.003)	.003 (.004)
Average trips per month	-.049*** (.015)	-.007 (.031)
Log monthly salary	.090** (.042)	.026 (.057)
Observations	4,510	2,079
F-statistic on joint test of all contents/truck characteristics	24.61	2.90
p-value of joint test of all contents/truck characteristics	.00	.41
F-statistic on joint test of all driver characteristics	17.70	8.12
p-value of joint test of all driver characteristics	.01	.23

NOTE.—This table presents the results from estimating eq. (10), where there is one observation for each payment at a checkpoint, and checkpoint $\times$ direction $\times$ month fixed effects and hour of day fixed effects are included. Robust standard errors are in parentheses, adjusted simultaneously for clustering at both the checkpoint and trip levels.
* Significant at 10 percent.
** Significant at 5 percent.
*** Significant at 1 percent.

Table 6: Price discrimination.

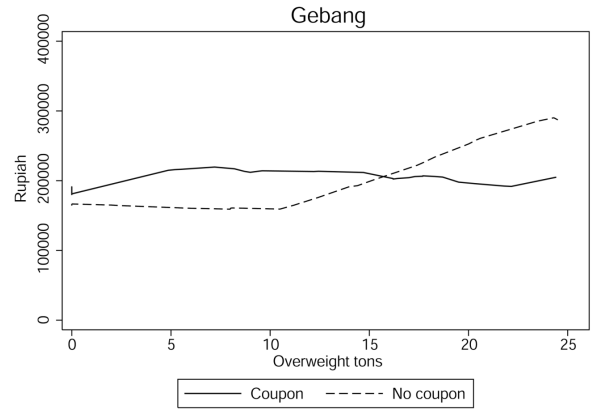


Figure 6: Payments at Gebang weigh station.

6.7 Table 7: Selection into the high fixed cost contract

Read:

- Table 7 studies which trucks choose the coupon contract at Gebang.
- Actual tons overweight is not statistically significant in the simple OLS specification.
- Driver salary strongly predicts coupon purchase.
- When tons overweight is instrumented with driver salary, the coefficient becomes larger and statistically significant.
- The authors interpret this as evidence that drivers may not know their exact weight before departure, so they choose the coupon based on expected or typical overweight status.

Economic message: The coupon system is a self-selection mechanism. Drivers with high expected overweight costs choose the high fixed cost, low marginal cost option.

TABLE 7
WHO SELECTS INTO A HIGH FIXED COST CONTRACT?

	OLS (1)	OLS (2)	IV (3)
Tons overweight	.003 (.010)		.039** (.019)
Log driver's monthly salary		.305** (.119)	
Constant	.495*** (.138)	-4.018** (1.760)	.074 (.210)
Observations	47	47	47
R^2	.00	.09	

NOTE.—This table reports the result of a linear probability model that estimates $\text{Coupon}_i = \alpha + X_i'\beta + \epsilon_i$, where Coupon is a dummy for whether a given trip i purchased the coupon and thus selected into the flat-price system. Each observation is a trip, and the sample is restricted to all trips on the Medan–Banda Aceh route headed in the direction of Banda Aceh, since coupons are for sale only on this route and only for this direction of travel. Columns 1 and 2 are estimated as linear probability models using OLS; col. 3 is estimated as a linear probability IV model using two-stage least squares, where the instrument is the log driver's monthly salary. Nonlinear probit and IV probit models produce qualitatively similar results. Robust standard errors are in parentheses. All columns drop two outliers: the highest observation on weight and the highest observation on salary. Including these observations strengthens the statistical significance of the results in cols. 2 and 3.

* Significant at 10 percent.

** Significant at 5 percent.

Table 7: Who selects into a high fixed cost contract?

7 Short summary

- The paper tests whether corrupt officials behave like profit-maximizing sellers.
- The setting is trucking in Aceh, Indonesia, where drivers pay many small illegal payments at checkpoints and weigh stations.
- The authors directly observe more than 6,000 illegal payments across 304 trips.
- A troop and police withdrawal from Aceh reduces the number of checkpoints, creating variation in market structure.
- When checkpoints in Aceh disappear, remaining checkpoints in North Sumatra raise prices.
- This endogenous price response means the total reduction in bribes is smaller than the mechanical reduction in checkpoints.
- Bribes are partly bargained: guns and more officers increase payments and negotiation probability.
- On the Meulaboh route, bribes rise later in the trip, consistent with downstream holdup.
- Officials price discriminate based on truck age, cargo value, cargo type, and driver characteristics.
- The Gebang weigh station uses a coupon system that looks like a menu of two-part tariffs.
- The policy lesson is that anticorruption efforts must account for how remaining corrupt agents change their prices.

8 Conclusion

8.1 Core conclusion

Olken and Barron show that corrupt officials can behave much like firms. In the Aceh trucking market, bribes respond to market structure, bargaining power, route position, and observable willingness-to-pay proxies. Corruption is illegal, but the pricing of corruption follows recognizable economic logic.

8.2 Economic intuition

A truck driver needs to pass every checkpoint to complete the trip. This creates a chain of monopolists. When there are many checkpoints, each one charges a smaller amount, but the total route price can be high. When some checkpoints disappear, remaining checkpoints have room to raise their prices. At the same time, late checkpoints can extract more surplus because the driver has already invested time and resources in the trip.

8.3 Incremental contribution revisited

The paper’s contribution is to bring industrial organization ideas directly into the empirical study of corruption. It is not just documenting that bribes exist. It shows that bribe pricing behaves like pricing in a market with monopoly power, incomplete contracting, bargaining, and price discrimination.

8.4 Policy implications

The findings complicate simple anticorruption policies. Removing some corrupt checkpoints may reduce total bribes, but remaining officials may raise prices and partially offset the gains. If corruption is centralized, disrupting coordination can sometimes raise bribes by turning one coordinated monopolist into many decentralized sellers. Effective anticorruption policy needs to consider both the number of bribe takers and the organization of the corruption market.

8.5 Limitations

The setting is highly specific: a post-conflict region, long-distance trucking routes, and police or military checkpoints. The sample is not a random sample of all trucks. The paper has strong evidence of price responses, but it cannot always determine whether pricing is centralized or decentralized. The sophisticated coupon contract is observed at one weigh station, so its generality is uncertain.

8.6 Takeaway

The enduring takeaway is that corruption has a market structure. Bribes are not only about morality, enforcement, or punishment; they are also prices set by agents with market power. Understanding those prices is essential for designing anticorruption policy.